



FLAGSHIP — TrustedAgentOps

Provenance, Evaluation & Observability for Agentic Data Systems on Microsoft Fabric + Foundry

A combined flagship project unifying TrustedAnswer (provenance + continuous eval for Fabric Data Agents) and AgentOps Black Box (unified OTel for multi-agent + MCP). One reference architecture. One demo. Three audiences.

1. The thesis in one sentence

You cannot govern what you cannot observe, and you cannot trust what you cannot reproduce. TrustedAgentOps makes every answer from every agent — single-agent, multi-agent, or Fabric Data Agent — reproducible, evaluable, and auditable on the Microsoft stack, without adding a third-party SaaS.

2. The compound problem

Enterprises deploying agents on Microsoft Fabric + Foundry face two converging gaps that are usually treated separately but are actually the same gap:

Gap	Manifests as
Data-side blind spot	Fabric Data Agent answers can't be reproduced; no ledger of generated SQL/KQL + rows + sensitivity labels; drift is silent.
Agent-side blind spot	Multi-agent + MCP toolchains span Agent Framework, Foundry, Copilot Studio with no unified trace; cost, identity, and tool-arguments are scattered.

A wrong answer to "what was Q1 claims loss in Texas?" today requires *two different forensic investigations* — one in Fabric, one in Foundry. **TrustedAgentOps unifies them on one trace ID.**

3. Five capabilities (the product surface)

- Unified trace fabric** — Every agent, tool, and Fabric data-agent call emits OTel GenAI spans into **one Application Insights** workspace, with W3C trace context propagated through MCP. Foundry Observability connected for the agent dashboard view.
- AnswerLedger** — Provenance store in KQL: `{trace_id, user, agent_id, prompt, generated_query, source_tables, sensitivity_labels, row_count, model, tokens, cost, eval_scores, red_team_flags}`.
- Continuous evaluation** — Golden-question harness running nightly via GitHub Actions / Azure DevOps; Foundry continuous evaluators (groundedness, intent-resolution, tool-call accuracy, retrieval F1); regression diff between versions.

4. **Trust & Cost dashboard** — One Real-Time Dashboard (KQL) + Azure Workbook: per-domain trust score, drift alarms, cost-per-conversation, agent-handoff Sankey, tool-call success matrix, sensitivity-label hit rate.
5. **Security overlay** — AI Red Teaming Agent in CI; Sentinel rules for answer-drift, anomalous tool-args, anomalous sensitivity-label access; Purview DSPM/DLP integration where available.

4. Reference architecture

```
flowchart TB
    subgraph Users
        U[Business user / app]
    end

    subgraph "Agent layer (Microsoft Agent Framework + Foundry)"
        ORC[Orchestrator Agent]
        SA[Specialist Agents]
        WRAP[Thin Foundry wrapper<br/>around Fabric Data Agent]
    end

    subgraph "Tool layer"
        MCPW[MCP Client Wrapper<br/>injects W3C traceparent]
        MCP[(MCP Tool Servers)]
        FDA[Fabric Data Agent<br/>Warehouse / KQL / Semantic Model]
    end

    subgraph "Telemetry & provenance"
        OTEL[Azure Monitor OTel Distro]
        AI[Application Insights<br/>unified trace store]
        ES[Fabric Eventstream]
        LEDGER[(KQL AnswerLedger)]
    end

    subgraph "Eval & governance"
        GH[GitHub Actions<br/>continuous eval]
        FEVAL[Foundry Continuous Eval<br/>+ Agent Evaluators]
        RT[AI Red Teaming Agent]
        PV[Purview DSPM / DLP]
        SEN[Microsoft Sentinel]
    end

    subgraph "Surfaces"
        DASH[Real-Time Trust Dashboard]
        WB[Azure Workbook<br/>Cost · Sankey · Diff]
        FND[Foundry Observability]
        SOC[SOC / Data Ops]
```

```

end

U --> ORC
ORC -->|invoke_agent| SA
ORC -->|invoke_agent| WRAP
SA --> MCPW
WRAP -->|OB0| FDA
MCPW --> MCP

ORC & SA & WRAP & MCPW -.OTel GenAI spans.-> OTEL
OTEL --> AI
WRAP -->|provenance event| ES --> LEDGER
AI --> LEDGER

GH --> FEVAL --> LEDGER
RT --> AI
PV -.labels.-> WRAP

LEDGER --> DASH
AI --> WB
AI --> FND
LEDGER & AI --> SEN --> SOC

```

5. Why combining is better than the sum

Standalone weakness	How the flagship fixes it
TrustedAnswer alone has no story for <i>multi-agent</i> or <i>MCP tool</i> calls preceding the Fabric query.	Unified trace fabric covers the full call chain.
AgentOps alone has no <i>data-layer provenance</i> — it sees "tool returned 1.4 KB" but not which rows or sensitivity labels.	AnswerLedger gives the data context.
Two dashboards, two eval pipelines, two SOC integrations.	One trace ID, one ledger, one set of Sentinel rules.
Hard to justify to a CISO as a <i>security</i> investment vs. an ops one.	Red-teaming + Purview + Sentinel make it a Zero-Trust-for-Agents story.

6. 4-week flagship build plan

Week	Theme	Concrete deliverables
Week 1 — Foundation	Trace fabric + 1 agent end-to-end	• Foundry project + Fabric workspace + Agent Framework skeleton (1 orchestrator + 1 specialist). • Azure Monitor OTel Distro emitting <code>invoke_agent</code> / <code>execute_task</code> spans to App Insights.

		• Thin Foundry wrapper around 1 Fabric Data Agent (Healthcare or Insurance from this repo) with OBO + sensitivity-label capture.
Week 2 — MCP + Ledger	Close the tool blind spot, build provenance store	• MCP client wrapper that injects <code>transparent</code> ; 1 MCP tool server (e.g., KQL query tool); tool spans visible as children of <code>invoke_agent</code> . • Fabric Eventstream → KQL <code>AnswerLedger</code> table; provenance events written from the wrapper. • v1 Real-Time Dashboard: volume, latency, cost, sensitivity-hit rate.
Week 3 — Continuous eval + cost	Make quality and cost measurable	• Golden question set generated from <code>data_agent_queries/</code> + production traces. • GitHub Action nightly running Foundry continuous evaluators; scores joined to ledger. • Azure Workbook: cost-per-conversation, agent-handoff Sankey, version-diff regression view. • Drift alarm rule (KQL) on canonical-question divergence.
Week 4 — Security overlay + demo	Zero-Trust-for-Agents story	• AI Red Teaming Agent scan integrated into CI; findings overlaid on traces. • Sentinel analytic rules: answer-drift, anomalous tool-call args, anomalous sensitivity-label access. • Recorded end-to-end demo: schema change → drift alarm → trace → SQL diff → ledger → SOC ticket. • Forum deck + 1-pager per persona (CTO, Data lead, CISO). • Repo published with Bicep/azd one-command deploy.

Stretch (week 5+)

- Add a second industry vertical from this repo to prove cross-domain reuse.
- Wire **Entra Agent ID** so every agent is a distinct principal (sets up a follow-up project from Idea 4).
- Front the MCP server with **APIM as AI Gateway** for rate limiting + content safety (sets up Idea 1).

7. Demo storyboard (the 5-minute forum demo)

1. Business user asks: *"Show me Q1 nursing-documentation burden in cardiology vs ICU."*
2. Orchestrator agent decomposes → specialist agent → Fabric Data Agent → KQL.
3. Live trace view: `execute_task` → `invoke_agent` → `tool.call` → KQL span — **one trace, all systems**.
4. Open the AnswerLedger row: prompt, generated KQL, source tables, sensitivity labels, row count, cost, groundedness score.
5. Push a schema change. Continuous eval fires overnight. Drift alarm on the dashboard. Sentinel ticket auto-created.

6. Show the same trace surfaced in Foundry Observability and the Azure Workbook cost-per-conversation tile.
7. Run the AI Red Teaming Agent live; show a prompt-injection attempt flagged and tied to the trace.

8. Audience-tuned talking points

- **CTO:** "Cost-per-answer and trust-per-domain become SLAs. Pilot-to-production blocker removed."
- **Data / AI lead:** "First reference for multi-agent + MCP + Fabric Data Agent observability — entirely native."
- **CISO:** "Every agent answer is a forensic record: identity, prompt, query, rows, label, eval score, red-team flag."

9. What's reused from this repo

- `datasets/` and `data_agent_queries/` — substrate for the demo and golden-question generation.
- `fabriciq-nurse-doc-burden-usecase/` — KQL dashboard + Eventstream patterns to clone for AnswerLedger.
- `cross_industry_notebooks/agent_instructions/` — seed Foundry agent prompts.

10. Success metrics for the prototype

Metric	Target by end of week 4
Trace coverage	100% of orchestrator → specialist → tool → Fabric calls in one App Insights workspace
Provenance coverage	100% of Fabric Data Agent answers written to AnswerLedger
Eval coverage	≥ 30 golden questions, nightly, 4 evaluators
MTTD for drift	< 24h via continuous eval; < 5 min via Sentinel rule
MTTR debugging	< 10 min for a wrong-answer incident, from alert to root cause